Scientific Progress Assessment: Independent PyLith Reproduction

Axial Modeling Project

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Assessment

The project has a bounded PyLith workflow and several working model components, but it has not reproduced Cabaniss et al.'s reported eruption threshold or timing. The principal goal remains an independent PyLith reproduction of the paper's deformation, pressure, stress, and failure results, using the paper and supplement as specifications and comparison targets. The observation workflow uses Center BPR records for the 1998 and 2011 event windows. Separate-station holdout analyses are outside the reproduction scope.

The evidence supports a partial software implementation, not a reproduced eruption prediction. Center event-window pressure fits include the observed eruption deflation and therefore diagnose the model retrospectively. The 12–14 MPa temperature-dependent eruption overpressure remains a comparison target; the current failure criterion does not produce a stable threshold in that range.

Implemented model and limits

The PyLith workflow represents a 6 by 3 by 1 km pressurized ellipsoid centered 1.6 km below the seafloor, with its long axis oriented N30°W. The 50 by 50 by 10 km domain is a project assumption because the paper does not report the box dimensions. Four host cases are implemented: elastic, constant-property Maxwell, temperature-dependent Maxwell, and temperature-dependent Maxwell with a hydrothermal conductivity change. PyLith writes displacement and stress fields; project code maps thermal properties to the mechanical mesh and postprocesses tensile failure and face-connected Mohr–Coulomb paths.

Several inputs remain unresolved in the written source. The Maxwell branch fractions and relaxation times are provisional, the printed temperature law conflicts with the stated brittle and ductile limits, and the friction-angle notation is ambiguous. Tensile strength, pore pressure, full-depth density, and some boundary conditions are also unspecified. The current failure screen uses an explicit diagnostic proxy and is not a plastic fault-slip model.

The pressure calibrations use a fixed base, while the paper specifies a Winkler foundation. The finite-spring diagnostic has not been made reliable enough to replace that boundary: it accepts an initial residual of 5.40×10^3 under a 1.00×10^4 tolerance, and tighter solver settings previously failed in KSP.

Measured implementation evidence

The values below are the bounded results recorded in `docs/run_log.md` and `docs/KNOWN_ISSUES.md`. They are checks on current code behavior, not evidence that the paper's result has been recovered.

Check	Recorded result	What it establishes
Elastic cavity smoke	2,499 tetrahedra; 0.296 m peak uplift.	The bounded PyLith solve writes finite displacement and stress fields.
Spherical Mogi response	At 2,995 elements, vector-field error is 44.9% and center error is 55.9%. A nearby refined case gives 42.7% and 54.2%.	Nonnested meshes do not establish convergence; the analytical response is not yet matched closely.
Thermal mesh sensitivity	Sampled temperatures differ by as much as 324 °C near the reservoir between bounded meshes.	The nonnested mesh sequence does not establish spatial convergence.
Finite-spring boundary	Spring prior is 32,373 Pa/m; initial residual is 5.40×10^3 under a 1.00×10^4 tolerance.	The static solve is not sufficiently converged to replace the production fixed base.
Failure screen	No connected reservoir-to-surface path at 12–14 MPa under the current strength proxy for either basal treatment.	The reported eruption threshold is not reproduced.

The historical calibration uses Center BPR data for the 1998 and 2011 eruption windows. Each window contains the eruption deflation; these fits cannot independently predict eruption timing. Eight available processed Center archive files yielded 133,149 on-hour observations, retained by deployment with source provenance and gaps. Their common datum and the paper's exact channel selection remain unverified. The coarse prototype cavity mesh has a 34.90% geometric volume deficit relative to the specified ellipsoid, so its computed volume change cannot support Fig. 4b. A continuous Center pressure history, converged pressure scale, and stable joint failure criterion are still needed.

Reproduction status by paper result

Result	Status	Remaining work
Deformation and pressure histories	Partial. Center event-window fits run; full-history pressure and volume histories are not reproduced.	Establish a converged Center compliance and build the continuous forcing history used by the paper.
Four rheologies and material fields	Partial. The four configurations and property-transfer code run.	Resolve the modulus law and Maxwell spectrum, then compare the spatial fields with the paper.
Stress and failure fields	Partial. Saved-stress processing and connectivity checks run.	Validate mesh and time-step behavior with defensible strength inputs and reproduce the reported fields and eruption times.
Numerical benchmarks	Incomplete. Mogi, thermal, mesh, and basal-spring checks expose substantial error or sensitivity.	Establish mesh and domain convergence; stabilize the Winkler boundary and compare the specified analytical and finite-element benchmarks.

The panel-by-panel record in `docs/figure_reproduction.md` describes the remaining figure work. In particular, reservoir volume history, paper-time failure snapshots, the full set of thermal-property panels, and several supplementary benchmark sweeps are not yet reproduced.

Work sequence

1. Resolve paper ambiguities where the article, supplement, or a defensible independent measurement provides an answer. Mark all other choices as assumptions.
2. Refine the ellipsoid mesh and domain until Center displacement and cavity-adjacent stress are stable. Improve the Mogi agreement and use exact Center coordinates for compliance sampling.
3. Stabilize and verify the finite Winkler boundary against its analytical and supplementary benchmarks, then use it in pressure calibration.
4. Build one continuous Center BPR forcing history and calculate pressure, volume change, stress, and failure through the reported eruption cycles.
5. Refine the time integration until reservoir tensile failure and the connected Mohr–Coulomb path produce a stable pressure threshold. Then compare the independently obtained threshold with 12–14 MPa and the paper's event timing.

Evidence limits

The numerical results in this assessment are taken from the repository's tracked run record; raw BPR archives, processed series, meshes, and simulation fields remain local and untracked. This report refresh did not run tests, simulations, or new calibrations. `make reproduction-status` lists unresolved source requirements; `make reproduce` stops without claiming a completed experiment. The source paper and supplement are specifications, while their reported values are comparison targets. No author code, COMSOL files, or author-generated model fields are used.